

MATH 708. SPRING 2026. HOMEWORK 4. DUE ON MARCH 6.

- Unless otherwise stated, the exercises are from Voisin's textbook.
- The homework problems may be submitted orally (during assigned office hours), in written/typed form, or as a combination of the two.
- You may submit any subset of problems totaling up to 100 points.
- If you submit more than 100 points worth, I will grade only the first 100 points you designate.
- Do not write anything on the cover sheet, as it will be used for grading. Bring it to office hours when you submit problems orally, and staple it to your written/typed solutions.

Problem 1. (5 points) Let E and F be complex vector bundles determined by transition cocycles

$\{(U_i, \varphi_{ij} : U_i \cap U_j \rightarrow GL(r, \mathbb{C}))\}$ and $\{(U_i, \psi_{ij} : U_i \cap U_j \rightarrow GL(r', \mathbb{C}))\}$, respectively, on a common open cover $\{U_i\}$.

Identify transition cocycles for:

- (1) the direct sum $E \oplus F$,
- (2) the tensor product $E \otimes F$,
- (3) the dual bundle $E^\vee$,
- (4) the determinant line bundle $\det(E)$.

Problem 2. (5 points) Show that any short exact sequence of holomorphic vector bundles

$$0 \rightarrow L \rightarrow E \rightarrow F \rightarrow 0,$$

where L is a line bundle, induces short exact sequences of the form

$$0 \rightarrow L \otimes \bigwedge^{k-1} F \rightarrow \bigwedge^k E \rightarrow \bigwedge^k F \rightarrow 0$$

for all $k \geq 1$.

Problem 3. (5 points) Show that for any holomorphic vector bundle E of rank r there exists a non-degenerate pairing

$$\bigwedge^k E \times \bigwedge^{r-k} E \longrightarrow \det(E).$$

Deduce from this the existence of a natural isomorphism of holomorphic vector bundles

$$\bigwedge^k E \cong \left(\bigwedge^{r-k} E \right)^* \otimes \det(E).$$

Problem 4. (5 points) Let E be a holomorphic vector bundle on a complex manifold X , and let $Y \subset X$ be a complex submanifold of codimension at least 2. Show that the restriction map on global sections

$$H^0(X, E) \longrightarrow H^0(X \setminus Y, E)$$

is bijective.

Problem 5. (5 points) Let L be a holomorphic line bundle on a compact complex manifold X . Show that L is trivial if and only if both L and its dual L^* admit nontrivial global sections.

Problem 6. (10 points) Let L_1 and L_2 be two holomorphic line bundles on a complex manifold X . Suppose that $Y \subset X$ is a submanifold of codimension at least 2 such that L_1 and L_2 are isomorphic on $X \setminus Y$. Prove that $L_1 \cong L_2$.

Problem 7. (10 points) Show that the line bundle $\mathcal{O}(E)$, dual to the line bundle $\mathcal{O}(-E)$ (introduced in class), of the exceptional divisor

$$E = \mathbb{P}(N_{Y/X})$$

of the blow-up $\text{Bl}_Y(X) \rightarrow X$ of a compact complex manifold X admits (up to scaling) only one global section.

Problem 8. (10 points) Consider the $\mathbb{Z}/2\mathbb{Z}$ -action $z \mapsto -z$ on $\mathbb{C}^2$, which has one fixed point and whose quotient $\mathbb{C}^2/\{\pm 1\}$ is not smooth. Show that this action lifts to a $\mathbb{Z}/2\mathbb{Z}$ -action on the blow-up $\text{Bl}_0(\mathbb{C}^2)$, and prove that the quotient $\text{Bl}_0(\mathbb{C}^2)/\{\pm 1\}$ is a complex manifold.

Problem 9. (10 points) Let $C \subset \mathbb{C}^2$ be the curve defined by

$$z_1^2 = z_2^2(z_2 + 1).$$

Show that the closure of $C \setminus \{0\}$ (viewed as a subset of $\mathbb{C}^2 \setminus \{0\} = \text{Bl}_0(\mathbb{C}^2) \setminus E$) in the blow-up $\text{Bl}_0(\mathbb{C}^2)$ is a smooth curve.

Problem 10. (5 points) Let $\mathbb{C}^n \subset \mathbb{C}^{n+1}$ be the standard inclusion

$$(z_0, \dots, z_{n-1}) \mapsto (z_0, \dots, z_{n-1}, 0),$$

and consider the induced inclusion $\mathbb{P}^{n-1} \subset \mathbb{P}^n$. Show that the restriction of the Fubini–Study Kähler form ω_{FS} on $\mathbb{P}^n$ to $\mathbb{P}^{n-1}$ is the Fubini–Study Kähler form on $\mathbb{P}^{n-1}$.

Problem 11. (10 points) Let $A \in GL(n+1, \mathbb{C})$ and denote the induced automorphism by $\tilde{A} : \mathbb{P}^n \rightarrow \mathbb{P}^n$. Show that $\tilde{A}^* \omega_{\text{FS}} = \omega_{\text{FS}}$ (Fubini–Study Kähler form) if and only if $A \in U(n+1)$.

Problem 12. (10 points) Prove that

$$\int_{\mathbb{P}^n} \omega_{\text{FS}}^n = 1.$$

Problem 13. (5 points) Show that S^2 is the only sphere that admits a Kähler structure.¹

Problem 14. Let $\mathbb{Z}$ act on $\mathbb{C}^n \setminus \{0\}$ by

$$(z_1, \dots, z_n) \mapsto (\lambda^k z_1, \dots, \lambda^k z_n), \quad k \in \mathbb{Z},$$

where $0 < \lambda < 1$.

¹It is known that the only spheres which admit almost complex structures are S^2 and S^6 (Borel–Serre, 1953). It is a famous open problem to prove that S^6 does not admit a complex structure.

- (1) (10 points) Show that this action is free and properly discontinuous, and that the quotient

$$X = (\mathbb{C}^n \setminus \{0\})/\mathbb{Z}$$

is a complex manifold diffeomorphic to $S^1 \times S^{2n-1}$.

- (2) (10 points) For $n = 1$, show that X is isomorphic to a complex torus $\mathbb{C}/\Gamma$, and determine the lattice Γ explicitly.
 (3) (10 points) Show that X is Kähler if and only if $n = 1$.
 (4) (10 points) Generalize the construction by considering the $\mathbb{Z}$ -action on $\mathbb{C}^n \setminus \{0\}$ given by

$$(z_1, \dots, z_n) \mapsto (\lambda_1^k z_1, \dots, \lambda_n^k z_n), \quad k \in \mathbb{Z},$$

where $0 < \lambda_i < 1$ for all i . Show that the quotient $(\mathbb{C}^n \setminus \{0\})/\mathbb{Z}$ is again diffeomorphic to $S^1 \times S^{2n-1}$.

Problem 15. Let G be the complex Lie group consisting of all matrices of the form

$$\begin{pmatrix} 1 & z_1 & z_2 \\ 0 & 1 & z_3 \\ 0 & 0 & 1 \end{pmatrix} \in GL(3, \mathbb{C}).$$

Clearly, G is biholomorphic (as a complex manifold, not as a complex Lie group) to $\mathbb{C}^3$. The group G (and every subgroup of it) acts on G by multiplication in $GL(3, \mathbb{C})$. Consider the subgroup $\Gamma := G \cap GL(3, \mathbb{Z} + i\mathbb{Z})$.

- (1) (10 points) Show the quotient

$$X := G/\Gamma$$

is a complex manifold of dimension 3.²

- (2) (10 points) Show that the first and third coordinates define a holomorphic map

$$f : X \rightarrow \mathbb{C}/(\mathbb{Z} + i\mathbb{Z}) \times \mathbb{C}/(\mathbb{Z} + i\mathbb{Z})$$

with fibres isomorphic to $\mathbb{C}/(\mathbb{Z} + i\mathbb{Z})$.

- (3) (10 points) Show that X has a trivial holomorphic tangent bundle.

² X is not a Kähler manifold, but this is harder to prove than in Problem 14.